

CIS1300 Assignment 2 — StreakSafe

- This assignment is worth 4% of your final grade (see Section 2.0 for grading details).

Learning Outcomes

- Use basic C instructions, decision making statements, and loops
 - Read requirements carefully and follow them when writing code
 - Use problem-solving skills to solve the given problem
-

1.0 Background

A student productivity app called **StreakSafe** summarizes a learner's week using:

- study minutes
- breaks taken
- sleep hours

The app converts these real inputs into a **weekly score out of 100** and a letter grade. You will write a C program that processes **multiple students** (one at a time), prints a report for each, and prints the class average score at the end.

2.0 Important Rules

1. Name your C program file as `lastnameFirstnameA2.c` (**for example**, if first name is "Ritu" and last name is "Doe", then the file should be named as `doeRituA2.c`)
2. **NO arrays are allowed** in this assignment.
 - You must use loops and running totals (accumulators).
3. All outputs must show **2 digits after the decimal**.
4. Your program must compile with: `-std=c99 -Wall` with **no warnings**.
5. You may assume users enter values in the expected format.
6. **All prompts and outputs must exactly match the format shown in testScenariosA2. If there is a discrepancy, follow testScenariosA2.**
7. **This assignment will be graded in two parts:** 90% based on your submitted program and 10% based on a quiz about your submission, to be completed after the assignment due date. More information about the quiz will be provided later.

3.0 Program Flow (Required)

1. Print exactly:
Welcome to STREAKsafe admin page
2. For **Student# 1**, read inputs, compute the weekly score, print the report.
3. Ask whether to continue:
Would you like to continue: ('y' to continue, 'n' to exit:)

4. If 'y', process the next student (for example, Student#2)
5. When the user enters 'n', print:
Average weekly score of X students = YY.YY%

4.0 Inputs Per Student (day-by-day input)

4.1 Study minutes (7 days)

For each day (1 to 7), prompt exactly:

- Day <k> study minutes:

Meaning

- *study minutes* = actual number of minutes studied that day (0 to 300 is reasonable – assume that the user enters a valid and reasonable value – you **do not** need to validate it)

4.2 Breaks taken (7 days)

For each day (1 to 7), prompt:

- Day <k> breaks taken (0 to 10):

A “break” means a real pause from studying (walk, snack, stretch, etc.).

4.3 Sleep hours (7 nights)

For each day (1 to 7), prompt:

- Night <k> sleep hours (0.0 to 12.0):

Note: assume that the user enters a value between 0 and 12 – you **do not** need to validate it)

4.4 Exam week mode

Ask once per student:

- Is this an exam week? ('y' or 'n'):

This changes the weights (explained below).

5.0 Converting Real Inputs to Scores

You must convert daily inputs into **daily scores out of 100**, then average across the 7 days.

5.1 Study score conversion (minutes → score out of 100)

Let m be the day's study minutes entered.

- Cap minutes at **180** (3 hours):
 - If $m > 180$, treat it as 180.
- Daily study score:
 - $\text{studyScore} = (m / 180.0) * 100$

Examples:

- 0 min $\rightarrow$ 0
- 90 min $\rightarrow$ 50
- 180 min $\rightarrow$ 100
- 250 min $\rightarrow$ 100

5.2 Break score conversion (break count $\rightarrow$ score out of 100)

Let b be breaks taken that day:

- If b is 2, 3, or 4 $\rightarrow$ 100
- If b is 1 or 5 $\rightarrow$ 70
- If b is 0 or 6 $\rightarrow$ 40
- If $b \geq 7 \rightarrow$ 10

5.3 Sleep score conversion (hours $\rightarrow$ score out of 100)

Let h be sleep hours that night:

- If $7.0 \leq h \leq 9.0 \rightarrow$ 100
- If $6.0 \leq h < 7.0$ OR $9.0 < h \leq 10.0 \rightarrow$ 70
- If $5.0 \leq h < 6.0$ OR $10.0 < h \leq 11.0 \rightarrow$ 40
- Otherwise $\rightarrow$ 10

6.0 Required Checks (Apply in this order)

Check1: Week Freeze

If study minutes (prompted by section 4.1) for all 7 days were logged as 0, then all other contributions (break and sleep) are automatically frozen to 0, the overall weekly score is assigned as 0, and letter grade assigned is F.

- Study contribution = 0.00
- Break contribution = 0.00
- Sleep contribution = 0.00
- Overall weekly score = **0.00**
- Letter grade = **F**

If Week Freeze applies, do not prompt for breaks, sleep, or exam week for that student, and do not print the weight line. See `testScenariosA2` for an example.

Check2: Exam week weights

If exam week is 'y':

- Study weight = 50
- Break weight = 10
- Sleep weight = 40

Otherwise (normal week):

- Study weight = 40
- Break weight = 20
- Sleep weight = 40

Print exactly one of the following lines:

- If exam week print: **Weights: study = 50, breaks = 10, sleep = 40**
- If normal week print: **Weights: study = 40, breaks = 20, sleep = 40**

Check3: Wellness Gate (sleep requirement to pass)

Let `sleepContribution` be the sleep portion out of its sleep weight.

If `sleepContribution < (sleepWeight / 2.0)`

Then the student's final grade must be **F**, even if the overall weekly score is ≥ 50 .

(Example: `sleepWeight=40` $\rightarrow$ must have `sleepContribution` ≥ 20.00 to pass.)

7.0 Computing the Weekly Score

After you compute:

- `avgStudyScore` (0–100)
- `avgBreakScore` (0–100)
- `avgSleepScore` (0–100)

Convert each to its weighted contribution:

- `studyContribution = (avgStudyScore / 100.0) * studyWeight`
- `breakContribution = (avgBreakScore / 100.0) * breakWeight`
- `sleepContribution = (avgSleepScore / 100.0) * sleepWeight`

Then calculate the overall weekly score:

- `OverallWeeklyScore = studyContribution + breakContribution + sleepContribution`

8.0 Letter Grade Mapping

- A: 80–100
- B: 70–79.99
- C: 60–69.99
- D: 50–59.99
- F: 0–49.99

9.0 Output Format (Required)

You must print:

1. The three category summaries:

- Overall study score (<studyWeight>) = xx.xx / <studyWeight>
- Overall break score (<breakWeight>) = xx.xx / <breakWeight>
- Overall sleep score (<sleepWeight>) = xx.xx / <sleepWeight>

2. Then the final breakdown exactly like shown below:

```
Overall weekly score breakdown
*****
Study   = xx.xx
Breaks  = xx.xx
Sleep   = xx.xx
*****
Your overall weekly score = yy.yy%
Overall grade = Z
```

Section 10.0 shows a sample run for 4 students.

10.0 Sample Run (4 students)

Given in file testScenariosA2

11.0 Dos and Don'ts

- You are **not allowed** to use arrays in A2.
- You **are allowed** to use loops and running totals
- You will likely use:
 - for loops for 7 days
 - if / else if / else for score conversion rules
 - printf formatting with `%.2f`

- %% for printing the percentage sign %

12.0 Program Submission

You must submit 1 program file – name it `lastnameFirstnameA2.c`

Incorrect file names will result in penalty.

12.1 Program Expectations

Your program is expected to follow the outlined information exactly. Failure to do so will result in deductions to your assignment grade.

- The program files you submit MUST compile with NO warnings and errors using the flags - std=c99 -Wall.
- **Programs that fail to compile will be given a mark of 0.**
- Programs that produce warnings upon compilation will receive a deduction of 1 mark for each type/category of warning.
- Penalties will occur for missing style, comments, header comments.
- Your program file must contain the header comment shown in section 12.4 with the underlined & bolded sections properly filled in.

12.2 List of “Do Not”s

- Do not use Global Variables
 - Using global variables will result in 0 in the assignment
- Do not use GOTO statements
 - Using Goto statements will result in 0 in the assignment

12.3 Academic Integrity and Use of AI

You may use course materials, notes, and the textbook to support your learning.

Generative AI tools (e.g., ChatGPT, Copilot, Claude) may be used as learning supports, including to help explain concepts, review terminology, or discuss approaches to improving requirement quality.

However:

- You cannot use AI tools to complete this assignment.
- You remain responsible for all submitted work.
- Your final submission must demonstrate your own understanding and critical thinking.

12.4 Header Comment

Use the templates given below for header comment – you must include them in your program (c) file that you submit.

/*!\ Note: The file name, student name and email ID must be changed per file and per student.

/*****chaturvediRituA2.c*****/

Student Name: **Ritu Chaturvedi** Email Id: **ritu**

Due Date: October ... Course Name: CIS 1300

I have exclusive control over this submission via my password.
By including this statement in this header comment, I certify that:

1) I have read and understood the University policy on academic integrity. 2) I have completed the Computing with Integrity Tutorial on Moodle; and 3) I have achieved at least 80% in the Computing with Integrity Self Test.

I assert that this work is my own. I have appropriately acknowledged any and all material that I have used, whether directly quoted or paraphrased. Furthermore, I certify that this assignment was prepared by me specifically for this course.

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